

SECURITY CLEARANCE: MANUAL-ONLY EXPORT

CNC MILLING PATH: 2D BASE TOPOLOGY (Z=0)

System Identification: AELV-1M-AF / S-M-Platform (Venturi Correction)

Author Designation: User

Material Target: PE-HD Plates (High-Density Polyethylene)

Units: Millimeters (mm)

1. Global Coordinate System Setup

- Origin Point

$$(X = 0, Y = 0)$$

: Center of the stern (rear of the boat).

- X-Axis:** Port/Starboard width (Total span:

$$-600 \text{ mm}$$

to

$$+600 \text{ mm}$$

).

- Y-Axis:** Stern to Bow length (Total span:

$$0 \text{ mm}$$

to

$$+1500 \text{ mm}$$

).

2. Venturi Channel Verification (Inner Walls)

The inner walls must create a passive hydrodynamic funnel to accelerate the inflow velocity

$$v_1$$

to the required activation velocity

$$v_2$$

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- Inlet Width (

$$Y = 1500$$

):

$$X_{right} - X_{left} = 320 - (-320) = 640 \text{ mm}$$

- Throat Width (

$$Y = 750$$

):

$$X_{right} - X_{left} = 200 - (-200) = 400 \text{ mm}$$

- Acceleration Ratio:

$$\frac{A_1}{A_2} = \frac{640 \text{ mm}}{400 \text{ mm}} = 1.6$$

3. CNC Toolpath: Starboard Hull (Right Pontoon)

Directive: Linear cuts only. No interpolation, no fillets, no radiused corners.

- **START / POINT 1 (Inner Stern):** $X = 200, Y = 0$
 - **LINE TO POINT 2 (Inner Throat):** $X = 200, Y = 750$
(Note: This creates the parallel ejection channel for the jet and retrieval cassette.)
 - **LINE TO POINT 3 (Inner Bow / Entry Wedge):** $X = 320, Y = 1500$
(Note: This diagonal cut forms the right side of the Venturi funnel.)
 - **LINE TO POINT 4 (Outer Bow Shoulder):** $X = 600, Y = 1200$
(Note: This creates the sharp hydrodynamic wave-piercing bow.)
 - **LINE TO POINT 5 (Outer Stern):** $X = 600, Y = 0$
(Note: Straight outer boundary wall.)
 - **LINE TO POINT 1 (Close Polygon):** $X = 200, Y = 0$
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4. CNC Toolpath: Port Hull (Left Pontoon)

Directive: Exact Y-Z plane mirror of the Starboard hull.

- **START / POINT 1 (Inner Stern):** $X = -200, Y = 0$
- **LINE TO POINT 2 (Inner Throat):** $X = -200, Y = 750$
- **LINE TO POINT 3 (Inner Bow / Entry Wedge):** $X = -320, Y = 1500$
- **LINE TO POINT 4 (Outer Bow Shoulder):** $X = -600, Y = 1200$
- **LINE TO POINT 5 (Outer Stern):** $X = -600, Y = 0$
- **LINE TO POINT 1 (Close Polygon):** $X = -200, Y = 0$

5. Assembly Directives

- **Vertical Extrusion:** These 2D shapes represent the absolute boundary. The hull is built straight up from these coordinates to

$$Z = 400 \text{ mm}$$

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- **Joining:** All PE-HD joints must be extrusion welded exactly on these vertices. Material-on-material, sharp transitions. Do not grind down the outer bow shoulder points.